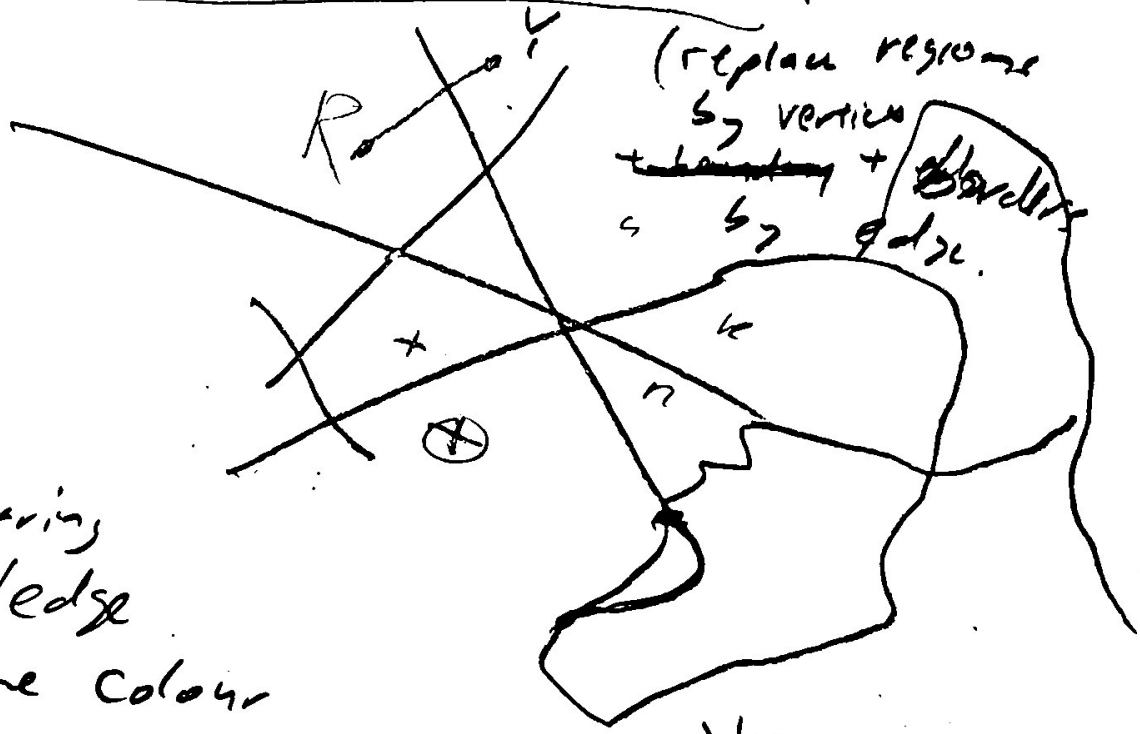
ECE 108

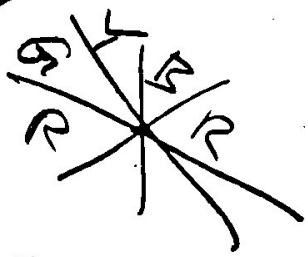
(1)

planar graph:

- You need at most 4 colors to colour a planar graph, so no two region sharing a "side"/edge are same colour

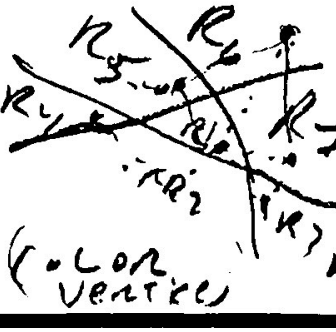
4 Colour Theorem





In graph, Hadwiger's conjecture. (H)

(H) ⇒ (4 Colour Thm)



Dual graph

Coloring connected vertices cannot have same color

$\Gamma = (V, E)$ $\quad V$ set

$E \subseteq V \times V$

$a, b \in V$

$(a, b) \in E$

Directed graph

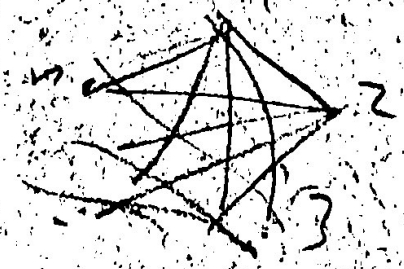
undirected graph

$(a, b) \in E \Rightarrow (b, a) \in E$

$\{a, b\} \in E \quad a \neq b$

← multigraph

K_n



complete graph

$E = (V \times V) \setminus \Delta$

$\Delta = \{(v, v) : v \in V\}$

K_n is a minor

Hadwiger if Γ requires t colors then K_t is a minor

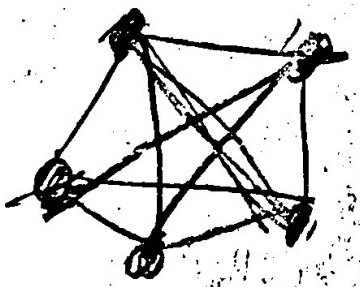
Suppose (H) is true.

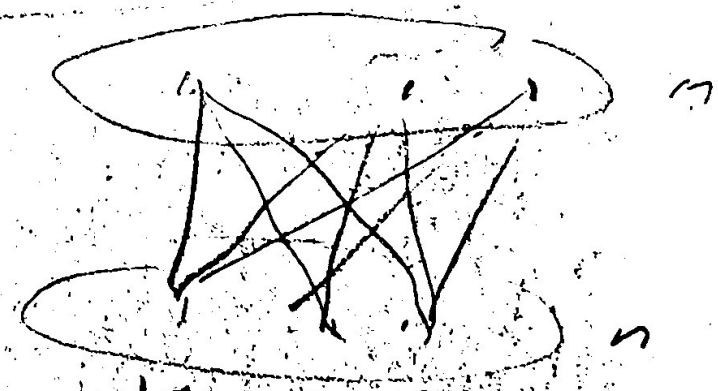
Suppose Γ is a planar graph.

Thm (Kuratowski)

$\Gamma = (V, E)$ is planar if and only if (iff)

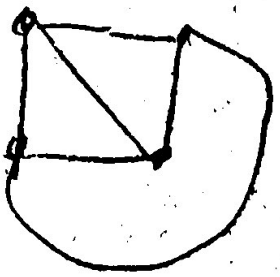
K_5 and $K_{3,3}$ are NOT minors of Γ.





$K_{m,n}$ complete bipartite graph

K_4 is planar.



If Γ is planar, K_5 is not a minor

By Hadiguer (H),

$\therefore \chi(\Gamma) < 5$ $\therefore$ 4 colors suffice for Γ.

chromatic number.

$\Gamma = (V, E)$ directed graph

For each edge $(a,b) \in E$

Define a $e_{a,b} : V \to V$

$$e_{a,b}(x) = \begin{cases} x & \text{if } x \neq a \\ b & \text{if } x = a \end{cases}$$ elementary collapsing

$e_{2,3}$:

1	2	3	4
↓	↓	↓	↓
1	3	3	4

$\mathcal{S}(\Gamma) = (V, S(\Gamma))$ Flow transformation semigroup (Rhodes)

set semigroup

$S(\Gamma) = \langle e_{a,b} : (a,b) \in E(\Gamma) \rangle$

"semigroup generated by"

$((V)\, e_{1,2})\, e_{3,2})\, e_{3,4}$

	1	2	3	4	composition mapping
$e_{1,2}$	↓	↓	↓	↓	
	2	2	3	4	
$e_{3,2}$	↓	↓	↓	↓	
	2	2	2	4	
$e_{3,4}$	↓	↓	↓	↓	
	2	2	2	4	

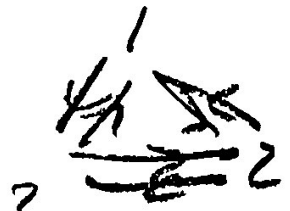
$e_{12} e_{32} e_{34} \in S(\Gamma)$

Thm Transformation Flow semigroup $(V, \mathcal{S}(\Gamma))$ determines Γ. if Γ is undirected graph

Γ

3 edges

K_3



6 edges.

$V = \{1, 2, 3\}$

$$(V, S(\Gamma)) = (V, S(K_3))$$

$$\langle e_{1,2}, e_{2,3}, e_{3,1} \rangle = S(\Gamma)$$

Find $e_{2,1} \in S(\Gamma)$

$e_{3,2}$

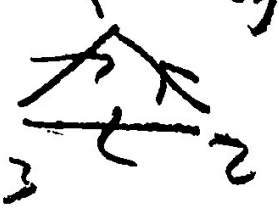
$e_{1,3}$

J. Rhodes

Application of Automata & Algebra to...

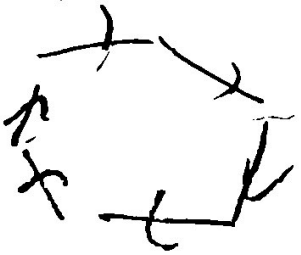
cycle

not cycle



If $(a,b) \in E$ is in a cycle in Γ, then $e_{a,b}$ & $e_{b,a} \in S(\Gamma)$

Γ

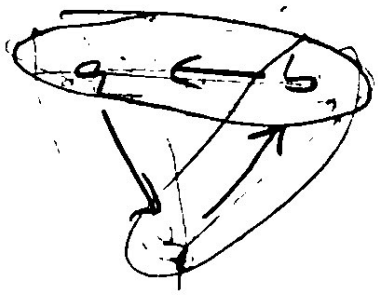


$\rightsquigarrow$

$S(\Gamma)$

$$e_{ab} = (e_{ba}\, e_{zb}\, e_{az})^{z}$$

$a \longrightarrow b$



$n = 3$

$$(\quad)\, e_{ba}\, e_{zb}\, e_{az}\, e_{bc}\, e_{zb}\, e_{az}$$

$e_{a_1 a_2}$